AST.H

```
#ifndef AST_H
#define AST_H

#include <stddef.h>

typedef enum {
    AST_FUNCTION,
    AST_PARAMETER,
    AST_CALL,
    AST_RETURN,
    AST_BINARY,
    AST_IDENTIFIER,
    AST_NUMBER
} ASTType;

typedef struct ASTNode {
    ASTType type;
    char *value;
    struct ASTNode *children;
    struct ASTNode *next;
} ASTNode;

/* AST construction */
ASTNode *create_node(ASTType type, const char *value);
void add_child(ASTNode *parent, ASTNode *child);

ASTNode *make_function(const char *name);
```

```
ASTNode *make_parameter(const char *name);
ASTNode *make_call(const char *name);
ASTNode *make_return(ASTNode *expr);
ASTNode *make_binary(const char *op,
                     ASTNode *left,
                     ASTNode *right);
ASTNode *make_identifier(const char *name);
ASTNode *make_number(const char *text);

/* AST utilities */
void free_ast(ASTNode *node);
void print_ast(const ASTNode *node, int depth);

#endif
```